

Nepal Import Review Draft for SunBridge Trading

1. Executive Summary

The supplied documents provide evidence of PV inverter testing and certification activities, but they do not appear to describe the same inverter model family.

The test report appears to cover CHISAGE CE-1P single-phase inverter models.

The certificate appears to cover Deye SUN G06P3 inverter models.

Several NEPQA-relevant standards were identified, including IEC 61727, IEC 62116, and IEC 62109-1.

However, the exact inverter model intended for import could not be confirmed from the provided documents.

Additional documentation may be required before a Nepal import review can be completed.

2. Documents Reviewed

- SGS IEC/EN 62109-1 Test Report
- SGS Certificate of Conformity

3. Product and Variant Review

The reviewed documents appear to describe different grid-connected photovoltaic inverter families.

SGS IEC/EN 62109-1 Test Report - CHISAGE CE-1P single-phase inverter models

- Product type: Single phase inverter
- IP rating: IP67

Models listed:

- CE-1P3001G-230-EU
- CE-1P5001G-230-EU
- CE-1P6001G-230-EU
- CE-1P8001G-230-EU

- CE-1P10001G-230-EU
- CE-1P13001G-230-EU
- CE-1P16001G-230-EU
- CE-1P18001G-230-EU
- CE-1P20001G-230-EU

SGS Certificate of Conformity - Deye SUN G06P3 inverter models

- Product type: Grid-connected PV Inverter
- IP rating: IP65

AM2 models:

- SUN-3K-G06P3-EU-AM2
- SUN-4K-G06P3-EU-AM2
- SUN-5K-G06P3-EU-AM2
- SUN-6K-G06P3-EU-AM2
- SUN-7K-G06P3-EU-AM2
- SUN-8K-G06P3-EU-AM2
- SUN-9K-G06P3-EU-AM2
- SUN-10K-G06P3-EU-AM2
- SUN-12K-G06P3-EU-AM2
- SUN-15K-G06P3-EU-AM2

AM2-P1 models:

- SUN-3K-G06P3-EU-AM2-P1
- SUN-4K-G06P3-EU-AM2-P1
- SUN-5K-G06P3-EU-AM2-P1
- SUN-6K-G06P3-EU-AM2-P1
- SUN-7K-G06P3-EU-AM2-P1
- SUN-8K-G06P3-EU-AM2-P1
- SUN-9K-G06P3-EU-AM2-P1
- SUN-10K-G06P3-EU-AM2-P1
- SUN-12K-G06P3-EU-AM2-P1
- SUN-15K-G06P3-EU-AM2-P1

4. Manufacturer and Certificate Information

- Manufacturer named in the test report: Zhejiang CHISAGE New Energy Technology Co., Ltd
- Applicant named in the test report: Zhejiang CHISAGE New Energy Technology Co., Ltd

- Factory named in the test report: NingBo Deye Inverter Technology Co., Ltd.
- Certificate holder named in the certificate: NingBo Deye Inverter Technology Co., Ltd.
- Certificate number: PCS-24-1022
- Test report number: GZES230100125901

5. Test and Standards Evidence

NEPQA-relevant standards identified:

- IEC 61727:2004
- IEC 62109-1:2010
- IEC 62116:2014

Other referenced standards identified:

- IEC 17065:2012
- IEC 60309
- IEC 60364-1
- IEC 60364-5-54
- IEC 60417
- IEC 60529
- IEC 60664
- IEC 60664-1
- IEC 60707
- IEC 60755
- IEC 60990
- IEC 62020
- IEC 62109
- IEC 62852

Standards by reviewed document:

- SGS IEC/EN 62109-1 Test Report: IEC 60309, IEC 60364-1, IEC 60364-5-54, IEC 60417, IEC 60529, IEC 60664, IEC 60664-1, IEC 60707, IEC 60755, IEC 60990, IEC 62020, IEC 62109, IEC 62109-1:2010, IEC 62852
- SGS Certificate of Conformity: IEC 17065:2012, IEC 61727:2004, IEC 62116:2014

6. Nepal Import Review Reference Items

The following review reference items had evidence in the supplied documents:

- IEC 61727:2004 certificate/test evidence should be provided.
- IEC 62116:2014 islanding prevention test evidence should be provided.
- IEC 62109-1:2010 safety evidence should be provided.
- Ingress protection should be at least IP65 according to IEC 60529.
- Inverter should have fan cooling or an appropriate heat sink to avoid excessive heating.
- Inverter label should include manufacturer name.
- Inverter label should include brand, model, and type.

7. Consistent Information Across Documents

The following information appears broadly consistent:

- Both documents relate to grid-connected photovoltaic inverters.
- Both documents originate from SGS certification or testing processes.
- Both documents reference IEC-based standards.
- Both documents relate to products associated with NingBo Deye Inverter Technology Co., Ltd.

8. Mismatches and Items Requiring Confirmation

- Model family mismatch: the test report appears to cover CE-1P series, while the certificate appears to cover SUN G06P3 series.
- Product type mismatch: the test report identifies Single phase inverter, while the certificate identifies Grid-connected PV Inverter.
- IP rating mismatch: the test report identifies IP67, while the certificate identifies IP65.
- Manufacturer and certificate identity should be confirmed: the test report names Zhejiang CHISAGE New Energy Technology Co., Ltd as manufacturer, while the certificate holder is NingBo Deye Inverter Technology Co., Ltd.
- Standards evidence is split across the document set. The certificate supports IEC 61727 and IEC 62116 evidence; the test report supports IEC 62109-1 evidence.
- The exact inverter model intended for Nepal import could not be confirmed from the supplied documents.

9. Additional Documents Recommended

- IEC 62891 MPPT efficiency evidence.

- IEC 62109-2 inverter safety evidence.
- Warranty agreement with the principal inverter manufacturer.
- Technical catalogue and datasheet for the exact imported model.
- Voltage, frequency, phase sequence, THD, flicker, DC injection, and power factor specifications for Nepal grid-code review.
- Efficiency information, including MPPT efficiency, inverter efficiency, Euro efficiency, and no-load loss.
- Data logging and external user-interface information.
- Protection-function documentation for DC reverse polarity, grid fault, lightning feeder protection, automatic wake-up, synchronization, and shutdown.
- Final product label or nameplate showing rated power, input/output voltage and frequency, maximum input voltage, MPPT range, and serial number.

10. Limitations

- This draft is based only on the supplied documents and identified facts available at the time of review.
- It does not certify compliance or make an import approval decision.
- Certification-body listing, testing scope, and current certificate status were not independently verified.
- The exact inverter model intended for import should be confirmed before relying on this review package.